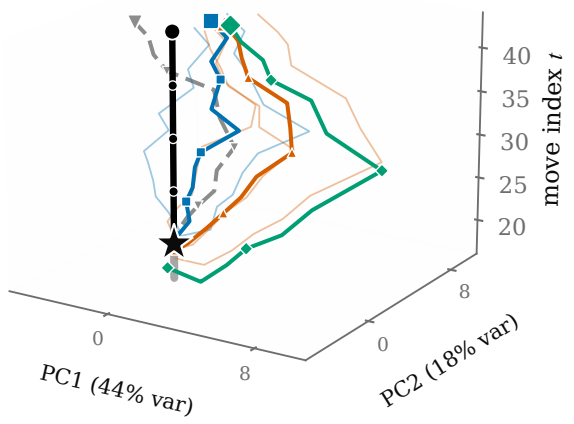
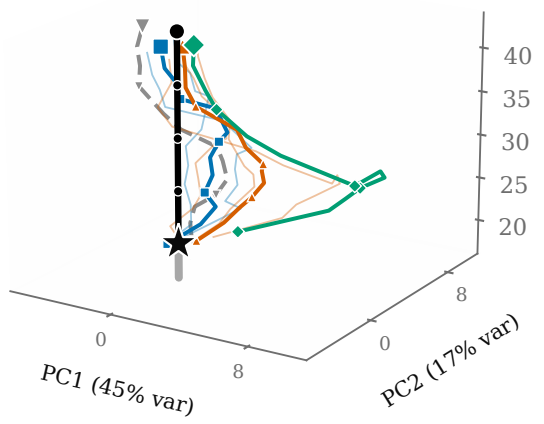
(a) where the branches go: the fork in each representation space, one path per rule

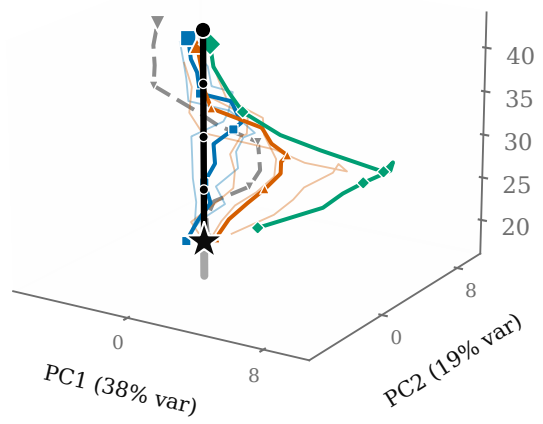
raw layer 5 (reference space)



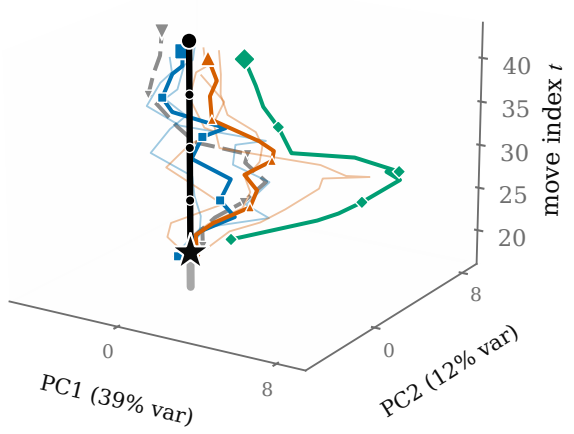
PAnneal (p-anneal, pen 0.042)



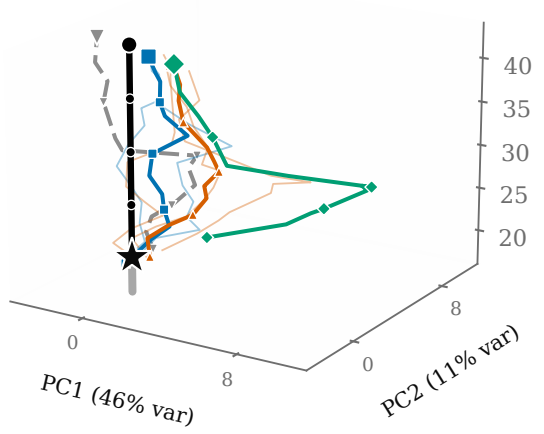
Standard (L1 0.059)



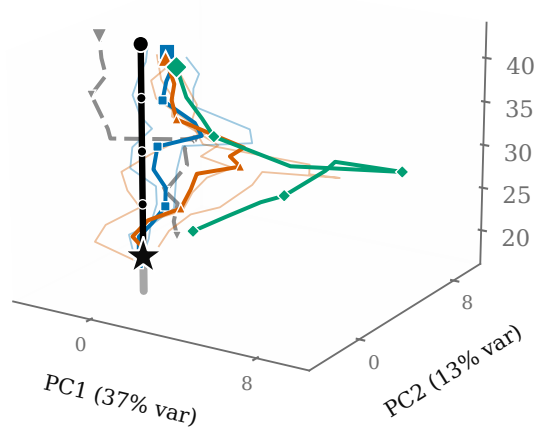
GatedSAE (L1 1.37)



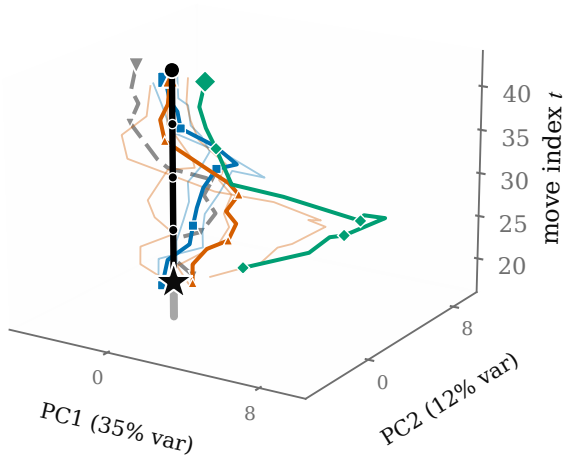
TopK (k=80)



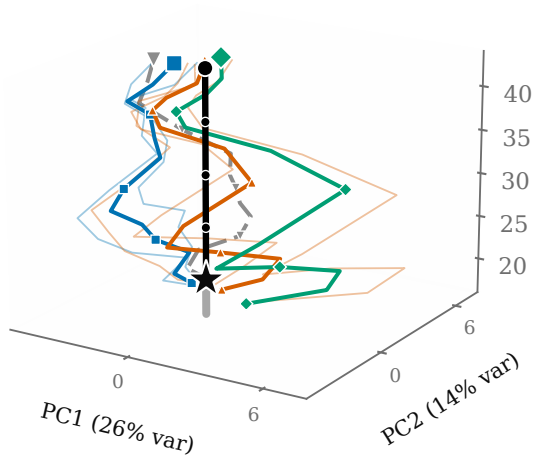
GatedAnneal (p-anneal, pen 0.955)



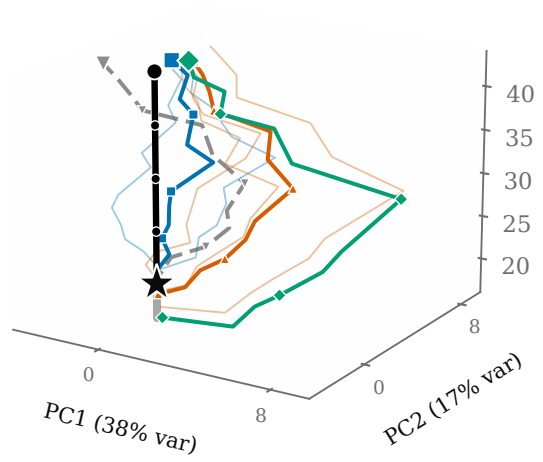
BatchTopK (k=80)



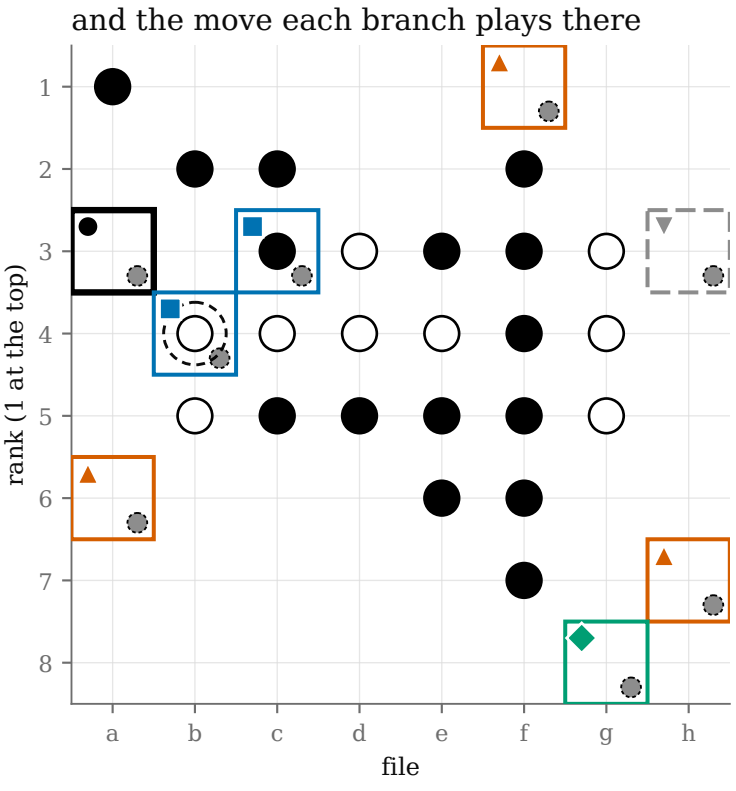
CosineSpaDE (γ=1.0)

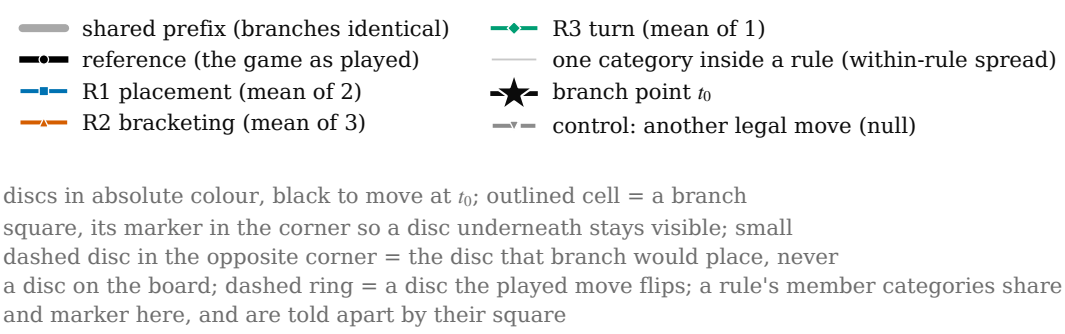


SpaDE (λ=1e-3, γ=1e-4)



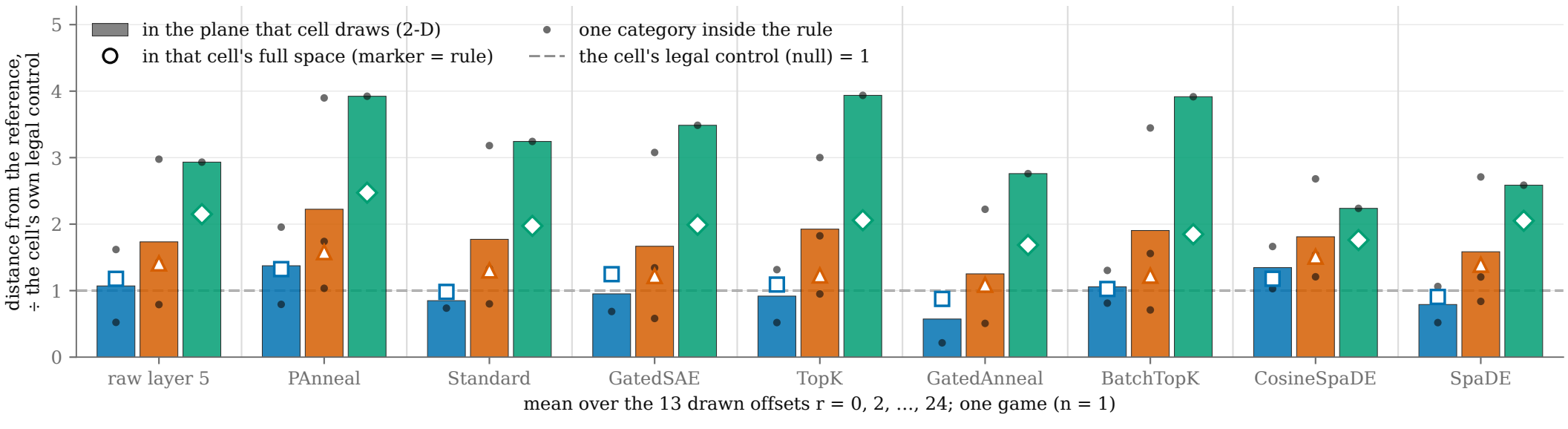
(b) what they are: the position at t_0 (Black to move), and the move each branch plays there





discs in absolute colour, black to move at t_0; outlined cell = a branch square, its marker in the corner so a disc underneath stays visible; small dashed disc in the opposite corner = the disc that branch would place, never a disc on the board; dashed ring = a disc the played move flips; a rule's member categories share one colour and marker here, and are told apart by their square

(c) how far they run: each rule's distance from the reference, in the plane its cell draws (bar) and in that cell's full space (open marker), as a multiple of the same cell's legal control



game 1235 is selected, not typical: the argmax over 1550 branch-complete games of its illegal branches' mutual separation (README)

R1 placement (2): square holds the mover's disc · square holds the opponent's disc

R2 bracketing (3): empty, touches no discs · empty, touches only the mover's discs · empty, touches opponent discs, no line closes

R3 turn (1): flips only for the opponent (turn confusion)